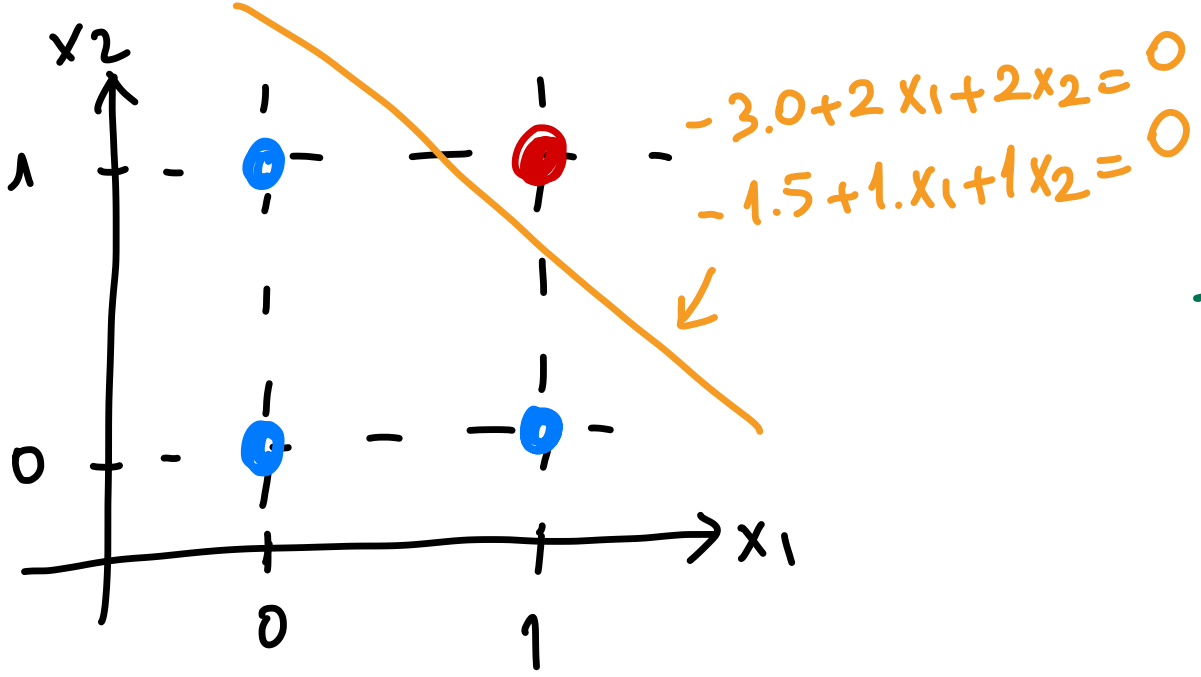
Boolean Functions

$x_1 \in \{0,1\}$ $x_2 \in \{0,1\}$

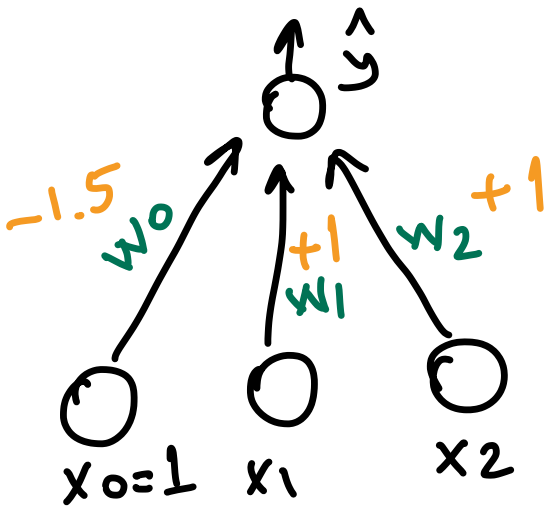
AND FUNCTION [x_1 AND x_2]

x_1	x_2	x_1 AND x_2
0	0	0
0	1	0
1	0	0
1	1	1



$$s(a) = \begin{cases} 1 & \text{if } a > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{y} = s(w_0 + w_1 \cdot x_1 + w_2 x_2)$$



x_1	x_2	$\hat{y}$
0	0	$s(-1.5+1x0+1x0) = 0$
0	1	$s(-1.5+1x0+1x1) = 0$
1	0	$s(-1.5+1x1+1x0) = 0$
1	1	$s(-1.5+1x1+1x1) = 1$

XOR FUNCTION [x_1 XOR x_2]

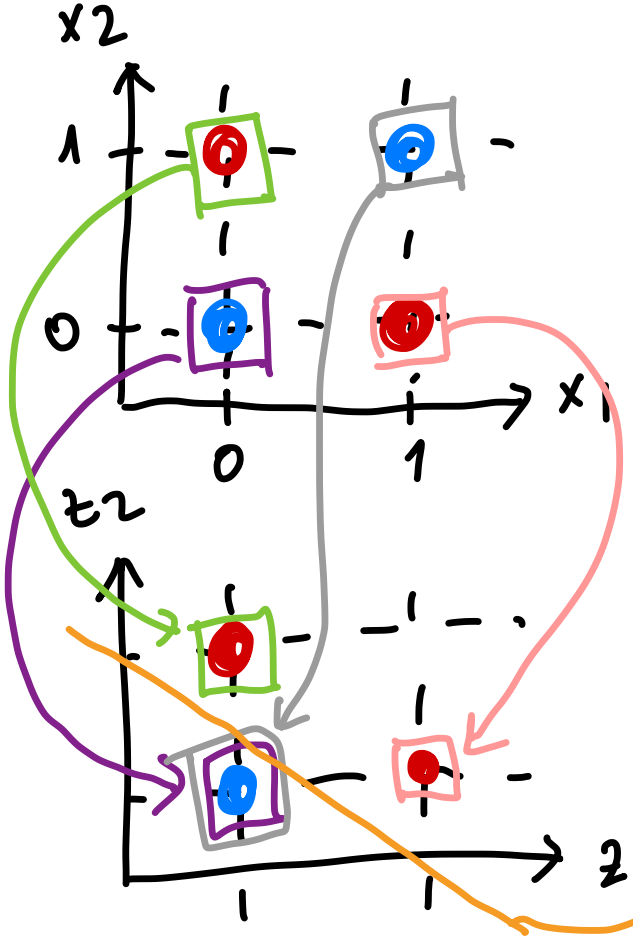
x_1	x_2	x_1 XOR x_2
0	0	0
0	1	1
1	0	1
1	1	0

$w_0 + w_1 x_0 + w_2 x_0 \leq 0$ (≤ 0, ≥ 0, ≥ 0)

$w_0 + w_1 x_0 + w_2 x_1 > 0$ (≤ 0, > 0)

$w_0 + w_1 x_1 + w_2 x_0 > 0$ (≤ 0, ≥ 0)

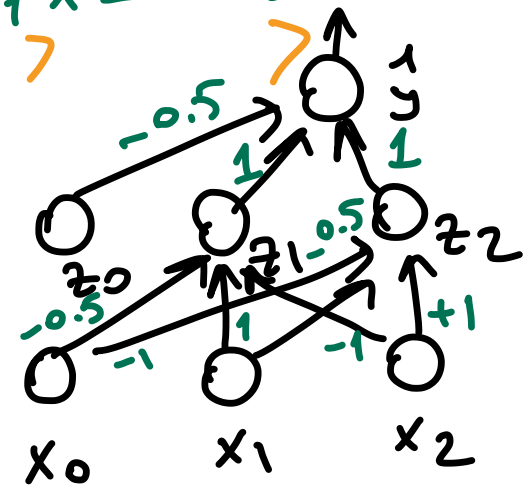
$w_0 + w_1 x_1 + w_2 x_1 \leq 0$ (≤ 0, $>$)



$z_1 = s(-0.5 + x_1 - x_2)$

$z_2 = s(-0.5 - x_1 + x_2)$

$\hat{y} = s(-0.5 + z_1 + z_2)$



x_1	x_2
0	0
0	1
1	0
1	1

z_1	z_2
0	0
0	1
1	0
0	0

$\hat{y}$
0
1
1
0

$f(x) = 3x+5$

$g(x) = x^2-2x+5$

$fog(x) = 3(x^2-2x+5)+5$

$= 3x^2-6x+20$

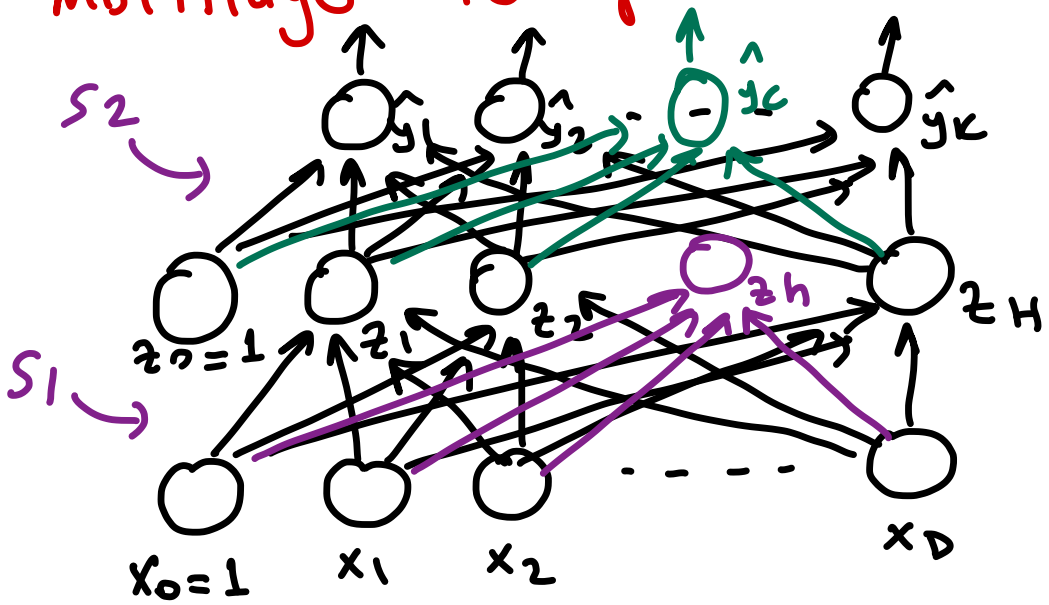
$f(x) = 3x+5$

$g(x) = 2x-4$

$fog(x) = 3(2x-4)+5$

$= 6x-7$

Multilayer Perceptrons



] output layer

] hidden/intermediate layer

] input layer

$(H+1) \times K$ V

$(D+1) \times H$ W

$$\#\text{ of model parameters} = \begin{cases} \text{single layer perceptron} & (D+1) \times K \\ \text{multilayer perceptron} & (D+1) \times H + (H+1) \times K \end{cases}$$



$(D+1) \times 1$ $(D \times 1) \times 1$

hidden nodes $\Rightarrow$ $z_h = s_1(w_h^T \cdot x)$

output nodes $\Rightarrow$ $\hat{y}_c = s_2(v_c^T \cdot z)$

Multiclass Classification

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in \mathbb{R}^D$ $\quad y_i \in \{1, 2, \dots, K\}$

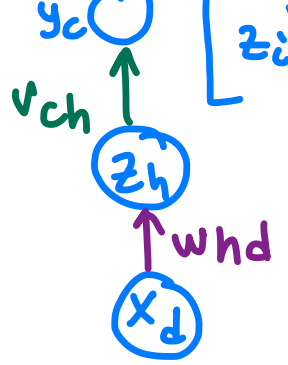
$S_1 \Rightarrow$ sigmoid (as an example)

$S_2 \Rightarrow$ softmax

$$z_{ih} = \text{sigmoid}(w_h^T . x_i)$$

$$\hat{y}_{ic} = \text{softmax}(v_c^T . z_i)$$

$$z_i = \begin{bmatrix} 1 \\ z_{i1} \\ z_{i2} \\ \vdots \\ z_{iH} \end{bmatrix}$$



$$\text{Error}_i = -\sum_{c=1}^{K} y_{ic} \log(\hat{y}_{ic})$$

$$\frac{\partial \text{Error}_i}{\partial w_{hd}} = \frac{\partial \text{Error}_i}{\partial \hat{y}_{ic}} \frac{\partial \hat{y}_{ic}}{\partial z_{ih}} \cdot \frac{\partial z_{ih}}{\partial w_{hd}}$$

$$\frac{\partial \text{Error}_i}{\partial v_{ch}} = \frac{\partial \text{Error}_i}{\partial \hat{y}_{ic}} \cdot \frac{\partial \hat{y}_{ic}}{\partial v_{ch}}$$

Nonlinear Regression

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$ $\quad x_i \in \mathbb{R}^D$ $\quad y_i \in \mathbb{R}$

$s_1 \Rightarrow$ sigmoid (as an example)

$s_2 \Rightarrow$ unit

$$\hat{y}_i = v^T \cdot z_i \rightarrow \sum_{h=1}^{H} v_h z_{ih} + v_0 \cdot z_{i0}$$

$$z_{ih} = \text{sigmoid}(w_h^T \cdot x_i) \rightarrow \sum_{d=1}^{D} w_{hd} \cdot x_{id} + w_{h0} x_{i0}$$

$$\text{Error}_i = \frac{1}{2}(y_i - \hat{y}_i)^2$$

$$\frac{\partial \text{Error}_i}{\partial v_h} = \frac{\partial \left[\frac{1}{2}(y_i - \hat{y}_i)^2\right]}{\partial v_h} = \frac{1}{2} \cdot 2 \cdot (y_i - v^T \cdot z_i)(-z_{ih})$$

$$= -(y_i - \hat{y}_i) z_{ih}$$

$$\frac{\partial \text{Error}_i}{\partial w_{hd}} = \frac{\partial \text{Error}_i}{\partial \hat{y}_i} \frac{\partial \hat{y}_i}{\partial z_{ih}} \cdot \frac{\partial z_{ih}}{\partial w_{hd}}$$

$\frac{1}{2} \cdot 2 (y_i - \hat{y}_i)(-1)$ $\quad v_h$ $\quad z_{ih}(1 - z_{ih}) x_{id}$

$$\Delta v_h = \eta (y_i - \hat{y}_i) z_{ih}$$

$$\Delta w_{hd} = \eta (y_i - \hat{y}_i) \cdot v_h \cdot z_{ih} (1 - z_{ih}) \cdot x_{id}$$

Binary Classification

$s_1 \Rightarrow$ sigmoid (as on example)

$s_2 \Rightarrow$ sigmoid

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N} \qquad x_i \in \mathbb{R}^D \qquad y_i \in \{0, 1\}$$

$$\hat{y}_i = \text{sigmoid}(v^T z_i) \rightarrow \sum_{h=1}^{H} v_h . z_{ih} + v_0 . z_{i0}$$

$$z_{ih} = \text{sigmoid}(w_h^T . x_i) \rightarrow \sum_{d=1}^{D} w_{hd} . x_{id} + w_{h0} . x_{i0}$$

$$\partial Error_i = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

$$\frac{\partial Error_i}{\partial v_h} = \frac{\partial Error_i}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial v_h}$$

$$= -\left[y_i . \frac{1}{\hat{y}_i} + (1 - y_i) \frac{(-1)}{1 - \hat{y}_i}\right] \hat{y}_i (1 - \hat{y}_i) . z_{ih}$$

$$= -\left[y_i (1 - \hat{y}_i) - (1 - y_i) \hat{y}_i\right] . z_{ih}$$

$$= -\left[y_i - y_i \hat{y}_i - \hat{y}_i + y_i \hat{y}_i\right] . z_{ih} = -\left[y_i - \hat{y}_i\right] . z_{ih}$$

$$\frac{\partial Error_i}{\partial w_{hd}} = \frac{\partial Error_i}{\partial \hat{y}_i} \frac{\partial \hat{y}_i}{\partial z_{ih}} \frac{\partial z_{ih}}{\partial w_{hd}}$$

$$= -\left[y_i \frac{1}{\hat{y}_i} + (1-y_i)\frac{(-1)}{1-\hat{y}_i}\right] \hat{y}_i.(1-\hat{y}_i).v_h.z_{ih}[1-z_{ih}].x_{id}$$

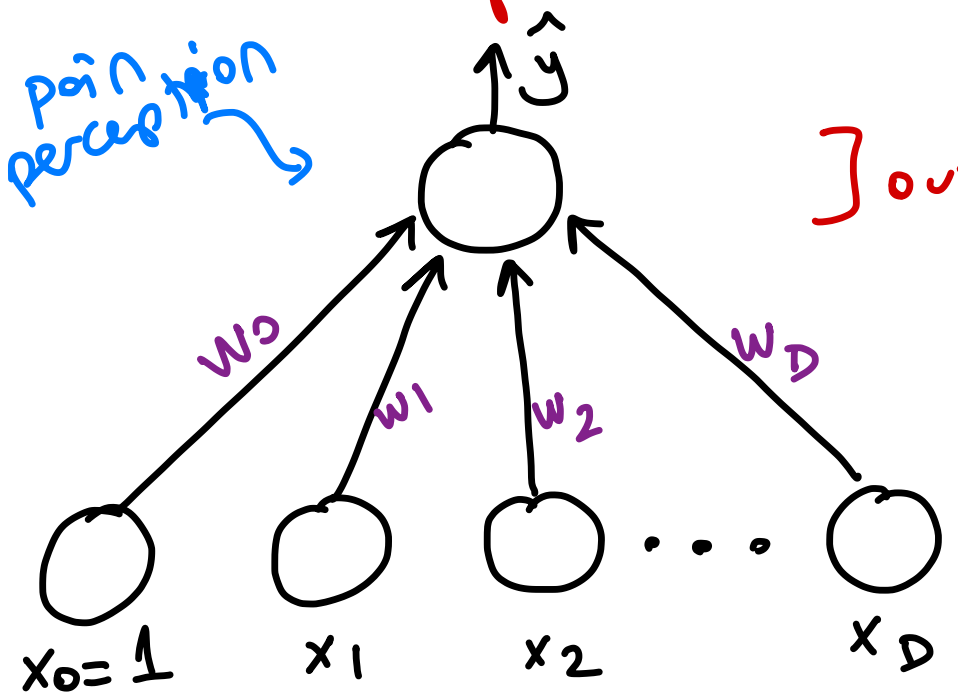
$$= -[y_i - y_i\hat{y}_i - \hat{y}_i + y_i\hat{y}_i].v_h.z_{ih}[1-z_{ih}].x_{id}$$

$$= -[y_i - \hat{y}_i].v_h.z_{ih}[1-z_{ih}].x_{id}$$

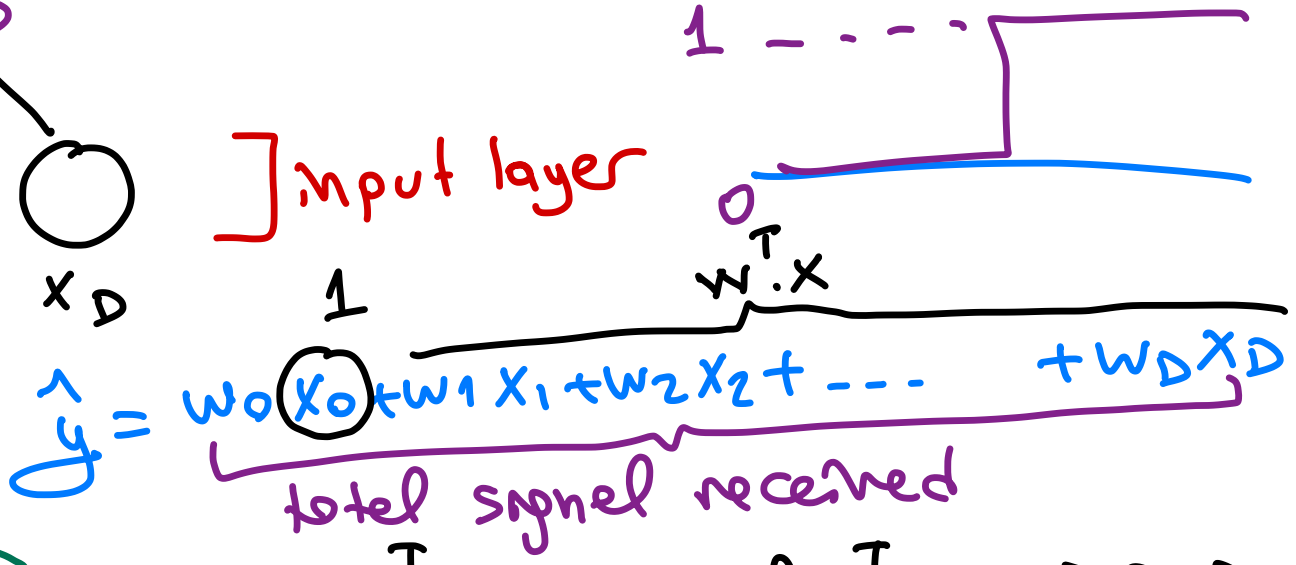
$$\Delta v_h = \eta (y_i - \hat{y}_i).z_{ih}$$

$$\Delta w_{hd} = \eta (y_i - \hat{y}_i).v_h.z_{ih}(1-z_{ih})x_{id}$$

The Perceptron



$w_0 + w_1 x_1 + w_2 x_2 + \dots + w_D x_D$?



$$\hat{y} = \underbrace{[w_1\ w_2\ \dots\ w_D]}_{1\times D}\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix}_{D\times 1} + \underbrace{w_0}_{1\times 1} = \underbrace{w_0 + w^T.x}_{w_0 + w^T.x}$$

$$= \underbrace{[w_0\ w_1\ w_2\ \dots\ w_D]}_{1\times(D+1)}\begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix}_{(D+1)\times 1} \quad \underbrace{\tilde{w}^T.\tilde{x}}$$

if $w^T.x + w_0 > 0 \Rightarrow$ pain

if $w^T.x + w_0 < 0 \Rightarrow$ no pain

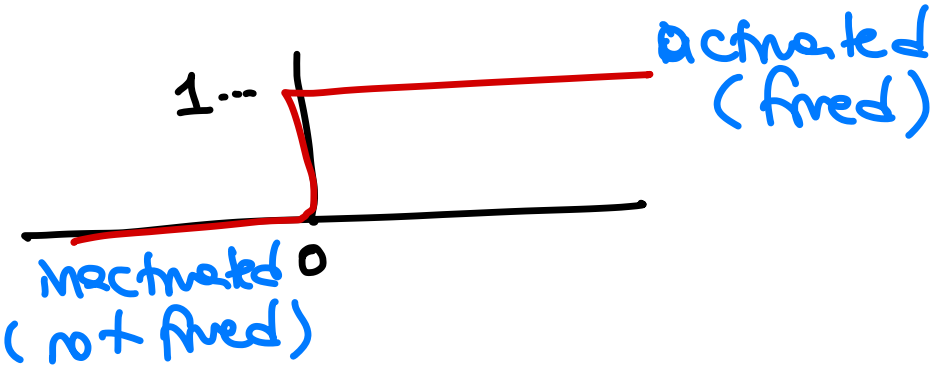
pain threshold

if $w^T.x > -w_0 \Rightarrow$ pain

if $w^T.x < -w_0 \Rightarrow$ no pain

Activation Function

$$s(a) = \begin{cases} 1 & \text{if } a > 0 \\ 0 & \text{otherwise} \end{cases}$$



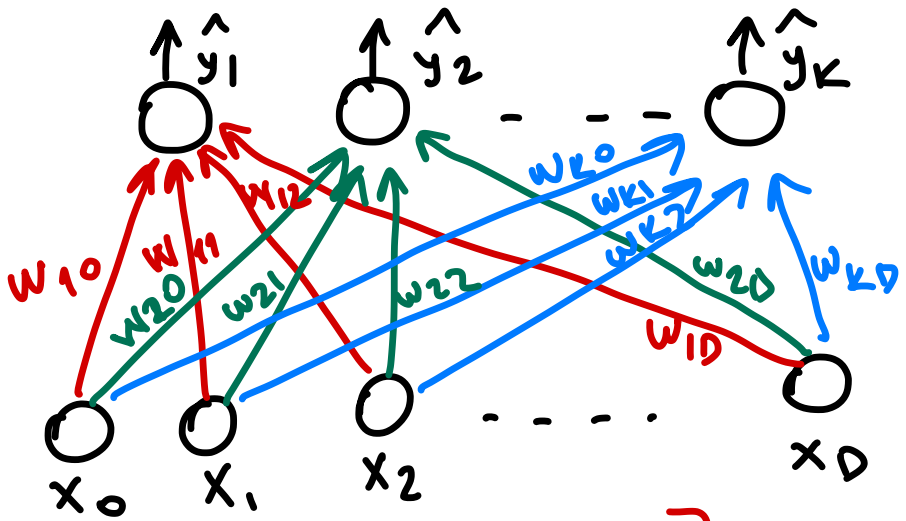
$$s(\tilde{w}^T \tilde{x}) = \begin{cases} 1 & \text{if } \tilde{w}^T \tilde{x} > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$s(\tilde{w}^T \tilde{x}) = \frac{1}{1 + \exp[-\tilde{w}^T \tilde{x}]}$$ } binary classification

↳ sigmoid activation

$$s(\tilde{w}^T \tilde{x}) = \tilde{w}^T \tilde{x}$$ } regression

↳ unit activation
linear activation



$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$$

$$x_i \in \mathbb{R}^D \qquad y_i \in \{1, 2, \ldots, K\}$$

$$\theta = \left\{ \tilde{w}_1 = \begin{bmatrix} w_{10} \\ w_{11} \\ w_{12} \\ \vdots \\ w_{1D} \end{bmatrix}, \tilde{w}_2 = \begin{bmatrix} w_{20} \\ w_{21} \\ w_{22} \\ \vdots \\ w_{2D} \end{bmatrix}, \ldots, \tilde{w}_K = \begin{bmatrix} w_{K0} \\ w_{K1} \\ w_{K2} \\ \vdots \\ w_{KD} \end{bmatrix} \right\}$$

model parameters

$$\hat{y}_1 = w_{10} + w_{11} \cdot x_1 + w_{12} x_2 + \cdots + w_{1D} x_D$$

$$\hat{y}_2 = w_{20} + w_{21} x_1 + w_{22} x_2 + \cdots + w_{2D} x_D$$

$$\hat{y}_K = w_{K0} + w_{K1} x_1 + w_{K2} x_2 + \cdots + w_{KD} x_D$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_K \end{bmatrix} = \begin{bmatrix} w_{10} & w_{11} & w_{12} & \cdots \\ w_{20} & w_{21} & w_{22} & \cdots \\ \vdots & \vdots & \vdots & \ddots \\ w_{K0} & w_{K1} & w_{K2} & \cdots \end{bmatrix} \begin{bmatrix} w_{1D} \\ w_{2D} \\ \vdots \\ w_{KD} \end{bmatrix} \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix}$$

$$\underbrace{\hat{y}}_{K \times 1} = \underbrace{\tilde{W}}_{K \times (D+1)} \cdot \underbrace{\tilde{x}}_{(D+1) \times 1}$$

$$\hat{y}_c = \sum_{d=1}^{D} w_{cd} \cdot x_d + w_{c0} = \tilde{w}_c^T \cdot \tilde{x} \quad \forall c$$

score function of class c
total signal received by neuron c

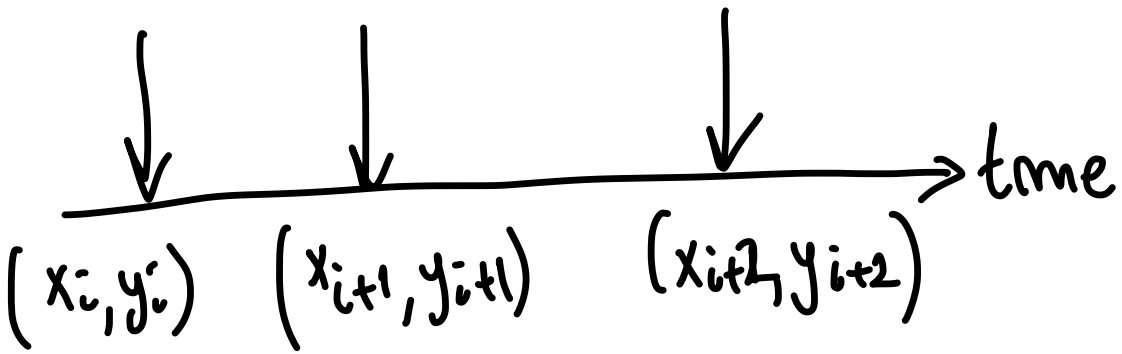
$$\hat{y}_c = \frac{\exp\left[\tilde{w}_c^T \cdot \tilde{x}\right]}{\sum_{d=1}^{K} \exp\left[\tilde{w}_d^T \cdot \tilde{x}\right]} \quad \forall c$$

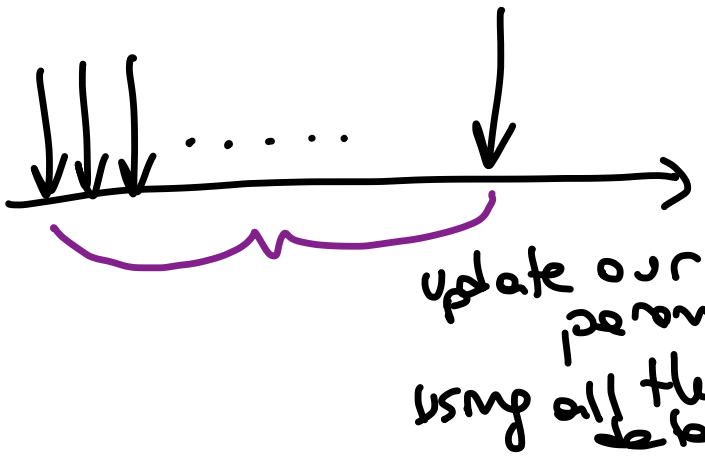
softmax activation

a new data point (x_{N+1}) choose $\hat{y} = \underset{c}{\operatorname{argmax}}\ \hat{y}_c$

LEARNING

Online Learning vs Batch Learning





Regression :

$$\frac{1}{2}\sum_{i=1}^{N}(y_i-\hat{y}_i)^2=\frac{\left[(y_1-\hat{y}_1)^2+(y_2-\hat{y}_2)^2+\dots+(y_N-\hat{y}_N)^2\right]}{2}$$

$\chi=\{(x_i,y_i)\}_{i=1}^{N}$

$x_i \in \mathbb{R}^D$

$y_i \in \mathbb{R}$

$$Error_i = \frac{1}{2}(y_i-\hat{y}_i)^2$$

$$\tilde{x}_i=\begin{bmatrix}1\\ x_i\end{bmatrix} \qquad \tilde{w}=\begin{bmatrix}w_0\\ w\end{bmatrix}$$

$$Error_i(\tilde{w}\,|\,x_i,y_i) = \frac{1}{2}(y_i-\hat{y}_i)^2$$

$$= \frac{1}{2}\left(y_i - s(\tilde{w}^T\tilde{x}_i)\right)^2$$

$$= \frac{1}{2}(y_i-\tilde{w}^T\tilde{x}_i)^2$$

$$\frac{\partial Error_i}{\partial \tilde{w}} = \frac{1}{2}\cdot 2\cdot(y_i-\underbrace{\tilde{w}^T\tilde{x}_i}_{\hat{y}_i})^{2-1}\cdot(-\tilde{x}_i)$$

$$= -(y_i-\hat{y}_i)\cdot\tilde{x}_i$$

$$\Delta\tilde{w} = -\eta\cdot-(y_i-\hat{y}_i)\cdot\tilde{x}_i = \eta\cdot(y_i-\hat{y}_i)\cdot\tilde{x}_i$$

Binary Classification

$$\sum_{i=1}^{N} \underbrace{-\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]}_{\text{Error}_i}$$

$$\chi = \{(x_i, y_i)\}_{i=1}^{N}$$

$$x_i \in \mathbb{R}^D$$

$$y_i \in \{0, 1\}$$

$$\text{Error}_i(\tilde{w} | x_i, y_i) = -\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

$$\hat{y}_i = s(\tilde{w}^T \tilde{x}_i) = \frac{1}{1+\exp\left[-\tilde{w}^T \tilde{x}_i\right]}$$

$$= -\left[y_i \log\left[\frac{1}{1+\exp\left[-\tilde{w}^T \cdot x_i\right]}\right] + (1-y_i)\log\left[1 - \frac{1}{1+\exp\left[-\tilde{w}^T \tilde{x}_i\right]}\right]\right]$$

$$\frac{\partial \text{Error}_i}{\partial \tilde{w}} = -(y_i - \hat{y}_i) \cdot \tilde{x}_i$$

$$\Delta \tilde{w} = -\eta \frac{\partial \text{Error}_i}{\partial w} = \eta (y_i - \hat{y}_i) \cdot \tilde{x}_i$$

Multi-class Classification

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$$

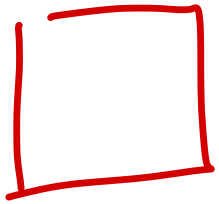
$$x_i \in \mathbb{R}^D$$

$$y_i \in \{1, 2, \ldots, K\}$$

$$\sum_{i=1}^{N} \underbrace{-\sum_{c=1}^{K} y_{ic} \log(\hat{y}_{ic})}_{\text{Error}_i}$$

$$\text{Error}_i(\{\tilde{w}_c\}_{c=1}^{K} \mid x_i, y_i) = -\sum_{c=1}^{K} y_{ic} \log(\hat{y}_{ic})$$

$$\hat{y}_{ic} = \frac{\exp[\tilde{w}_c^T \cdot \tilde{x}_i]}{\sum_{d=1}^{K} \exp[\tilde{w}_d^T \cdot \tilde{x}_i]}$$



CAT

CAT	DOG	HORSE
1	0	0

$$= -\sum_{c=1}^{K} y_{ic} \log\left[\frac{\exp[\tilde{w}_c^T \cdot \tilde{x}_i]}{\sum_{d=1}^{K} \exp[\tilde{w}_d^T \cdot x_i]}\right]$$

$$\frac{\partial \text{Error}_i}{\partial \tilde{w}_c} = -(y_{ic} - \hat{y}_{ic}) \cdot \tilde{x}_i$$

$$\Delta \tilde{w}_c = -\eta \frac{\partial \text{Error}_i}{\partial \tilde{w}_c} = \eta (y_{ic} - \hat{y}_{ic}) \cdot \tilde{x}_i$$